

Raphael Alcober

914-643-6450 | ralcober01@manhattan.edu | www.linkedin.com/in/ralphalc |
github.com/23alcor | 23alcor.github.io/Portfolio/
US Citizen

Education

- B.S. in Computer Engineering Manhattan University Expected May 2027

Research Publications

- Alcober, R., *Adversarially Robust Industrial Control Systems Using a Generative Purification Pipeline*, IEEE CCWC 2026
- Alcober, R., *Enhancing Robustness of Deep Learning Models in Power Systems Against Adversarial Attacks* *Innovations in Data Science and Analytics* (Chapter ID: DSA-23)

Projects

AI Automation Server

- Deployed a Python AI automation server integrating OpenAI API and Notion, automating task generation via API requests and JSON processing from personal devices.
- Built a FastAPI backend service to handle webhook events, process structured JSON payloads, and generate AI responses through prompt engineering and LLM API calls.
- Hosted on Raspberry Pi using Linux, systemctl, and Cloudflare tunnels to access outside of local network.

Work History

Engineering Co-Op at Consolidated Edison Failure Analysis, NY

Aug 2025 to Current

- Rewrote a decade-old internal data pipeline against a live SQL database, migrating off a deprecated frozen data source and eliminating a silent data-integrity bug that mislabeled records as they aged out.
- Packaged the internal machine-learning tool into a standalone Windows executable with PyInstaller, eliminating specific dependency-setup error on unconfigured machines and enabling one-click deployment for new co-ops. Used a Docker-based build environment for reproducibility.
- Built a Tkinter GUI front-end, giving non-technical engineers a point-and-click interface to run failure-classification workflows without using the command line.

Undergrad Researcher at Manhattan University, NY

Feb 2025 to Current

- Developed end-to-end data pipelines and model training workflows for Industrial Control System datasets using Python and PyTorch.

- Built scalable data preprocessing pipelines using Python, NumPy, and pandas to clean, filter, and transform large ICS datasets for downstream model training.
- Led a team of undergraduate researchers under faculty mentorship, incorporating iterative feedback and technical collaboration, using Git for version control, to refine project design and produce a published research paper.

Technical Skills

- **Languages:** Python, C++, SQL
- **Frameworks & Libraries:** PyTorch, NumPy, pandas, FastAPI
- **Developer Tools:** Git, Docker, Linux, PyInstaller, Tkinter, REST APIs

Leadership & Activities

Robotics Club at Manhattan University

- Worked in the AI Team in our overall team in the NYSID CREATE Assistive Technology competition; developed a real-time AI system to help assist a disabled employee in dish-return workloads.

Asian Student Association at Manhattan University

- Co-founded a club for students of any background to celebrate Asian culture and media; managed communications with members through email, meetings, and event announcements.